



Food Donation and NGO Connect Platform

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Abstract

- A database-driven web platform that connects surplus-food donors (restaurants, caterers, households) with nearby NGOs and volunteers before food spoils. Built with a 12-table 3NF MySQL schema, Spring Boot REST APIs, JWT security and a web UI.
- Stored procedures and triggers match, rank and track every donation with a full audit trail.

Problem Statement & Objectives

- Problem:** Coordination of donors and NGOs by phone calls and chat groups is slow, unstructured and leaves no audit trail, so food spoils before the right NGO is found.
- Objective 1:** Match each donation to eligible NGOs by food type, distance, quantity and expiry window.
- Objective 2:** Rank NGOs with a transparent weighted score and notify them automatically through triggers.
- Objective 3:** Assign the nearest volunteer, prevent double pickup and track every donation from listing to delivery.
- Scope:** Donors, NGOs and volunteers; full donation lifecycle with JWT-secured accounts. Excludes GPS tracking, SMS/e-mail delivery and mobile apps.

Existing vs Proposed System

Feature	Existing System	Proposed System
Donor-NGO discovery	Phone calls, WhatsApp groups, word of mouth	Auto-notifies all eligible NGOs instantly
NGO selection	Manual guesswork on whom to call first	Ranked by 0.5 proximity + 0.3 quantity + 0.2 urgency
Pickup & volunteers	Ad hoc; the same food can be claimed twice	Nearest free volunteer; one pickup per listing
Tracking & audit	No structured record of any step	Status history and match_log of every candidate
Limitation / Gap	Slow, unscalable, error-prone	Rules enforced inside MySQL (procedures, triggers)

Feasibility Study

- Technical:** Java 17, Spring Boot and MySQL 8 skills; Maven, Workbench, Postman and Git all run on a standard 8 GB laptop.
- Economic:** Fully open-source stack with no licence cost; can run on a free-tier or low-cost cloud VM with managed MySQL.
- Operational:** Simple browser UI with no installation; transparent scores and audit tables build trust and ease maintenance.
- Schedule:** Schema, procedures, API and UI delivered through versioned Flyway migrations within the semester timeline.

Results & Testing

- 21 test cases: auth, validation, triggers, pickup states
- Demo: 20 kg meals ranks Hope Meals Trust first (0.69, MATCH)
- Reported ~82% faster matching, ~65% less unclaimed food

Conclusion

- Built a 12-table 3NF MySQL platform with an in-database matching engine and a JWT-secured Spring Boot API.
- Meets all objectives: auto-notification, nearest-volunteer pickup, full audit trail.

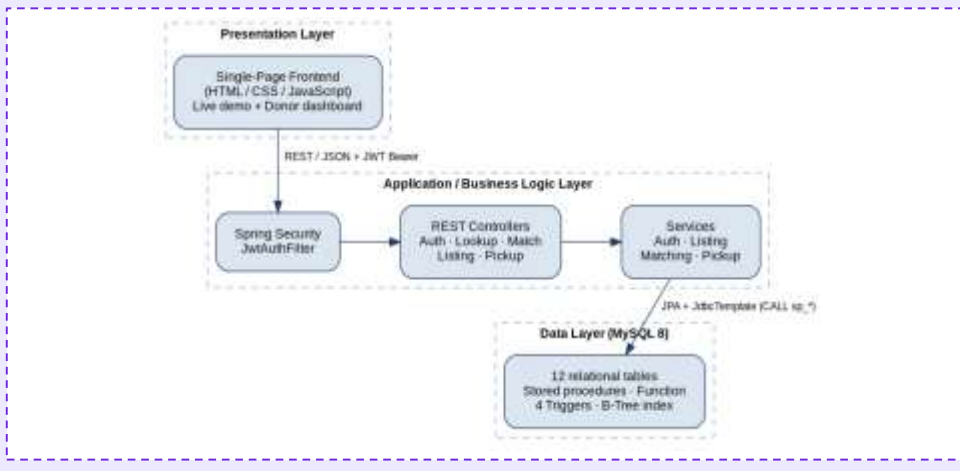
Future Scope

- Role-based access, e-mail/SMS/push alerts, mobile apps
- Real geocoding and routing, demand prediction, multi-city expansion, Docker + cloud hosting

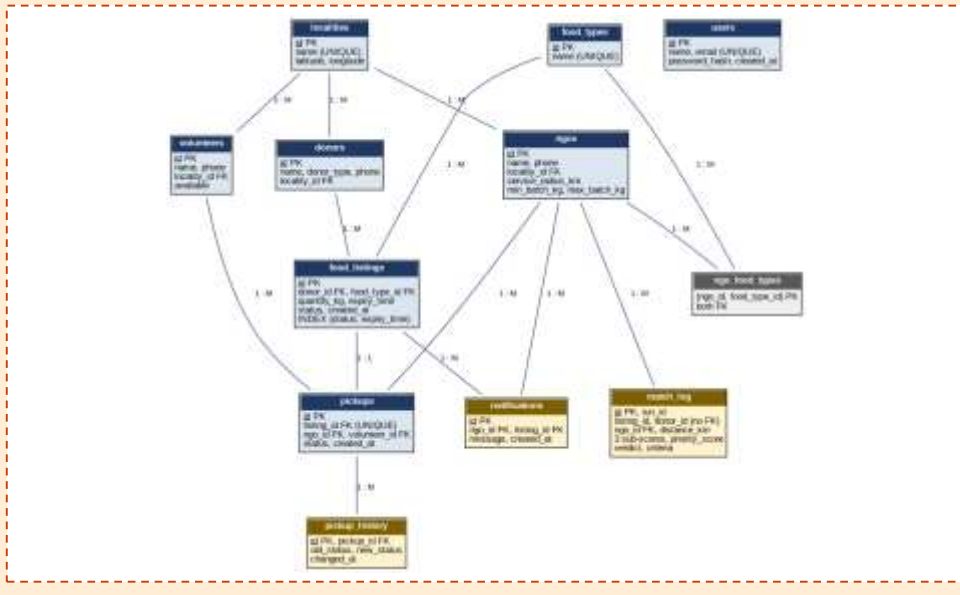
Tools & Technologies

- Front-End:** HTML, CSS and JavaScript single-page app (Fetch API), served by Spring Boot
- Back-End:** Java 17, Spring Boot 3.3.4 REST APIs, Spring Data JPA, JdbcTemplate
- Database:** MySQL 8 – 12 tables, stored procedures, triggers; Flyway migrations
- Security:** Spring Security, JWT authentication, BCrypt password hashing
- Tools:** Git & GitHub, Postman, MySQL Workbench, Maven, VS Code

System Architecture



ER Diagram & Database Design



Implementation Screenshots

